



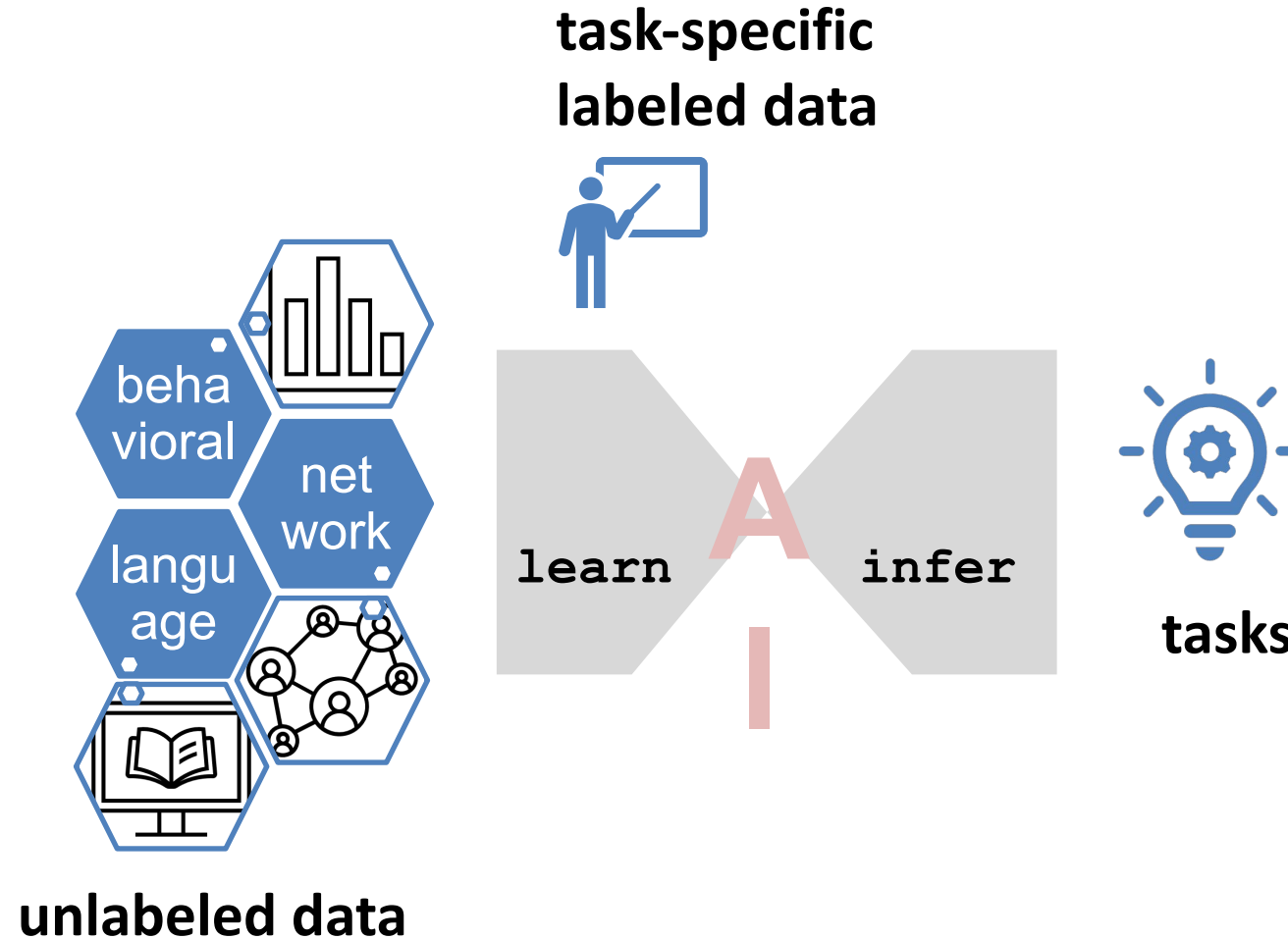
ALGORITHMIC BIAS AND FEEDBACK: BIAS ON THE WEB

Kristina Lerman

Spring 2025

DSCI 531

AI pipeline





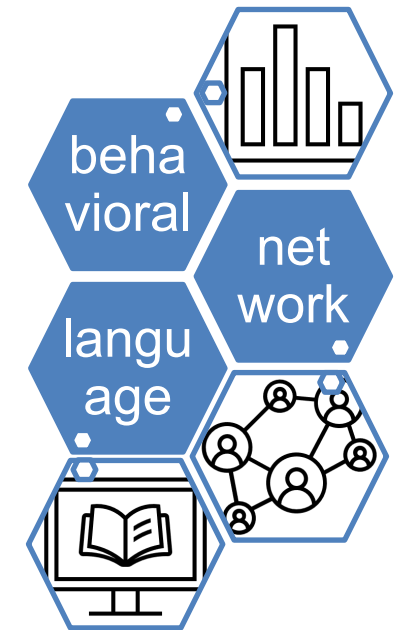
AI pipeline: feedback loops

Data is biased

- Cognitive biases
- Psychological biases: negativity, ingroup favoritism, ...
- Social biases: cultural biases & Stereotypes: body image, emotionality, moral values, ...
- Demographic: gender, ideology, ...

Data collection is biased

- Incomplete
- Non-representative
- ...



biased
unlabeled data

task-specific
labeled data

is noisy, biased,
subjective



learn

infer

algorithms are biased

- representation bias
- defining outcomes & features ...



tasks

Risks of bias

- AI models learned from biased data can unfairly discriminate
- People interact with AI outputs, they create more biased data
- This creates feedback loops that risk amplifying biases

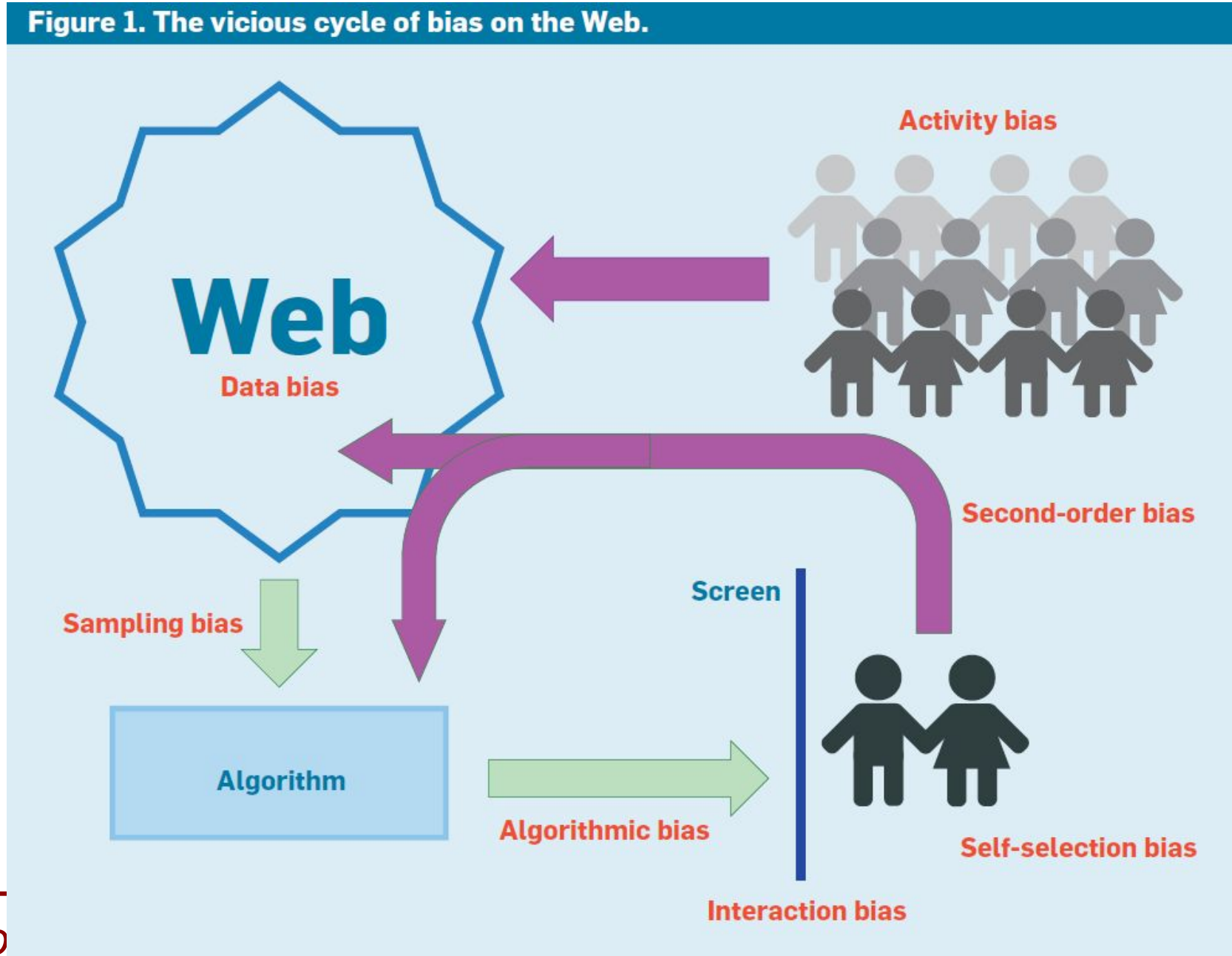
feedback loops amplify bias



Bias on the Web

[Baeza-Yates, R. (2018). Bias on the web. *Communications of the ACM*, 61(6), 54-61.]

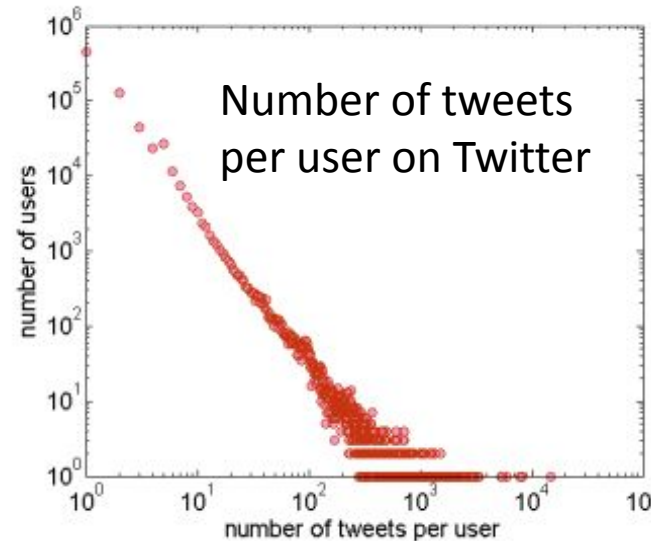
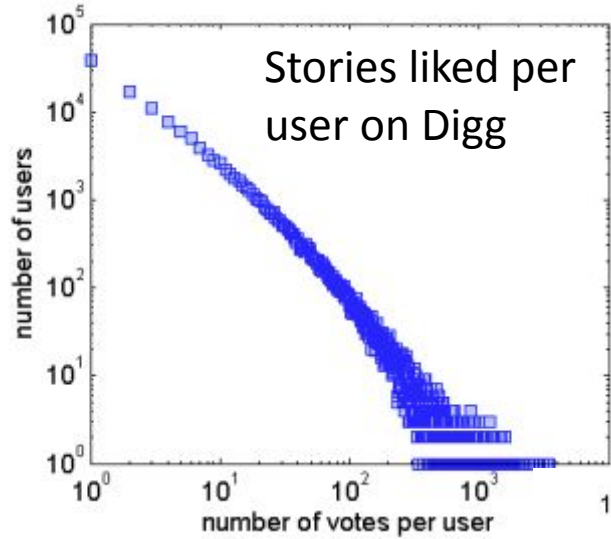
- **Any remedy for bias starts with awareness of its existence.**
- **Bias on the Web reflects biases within ourselves, manifested in subtler ways.**
- **We must consider and account for bias in the design of Web-based systems that truly address the needs of users.**



- Every decision and interaction creates an opportunity for bias to creep in
- Biases compound



Online engagement is highly heterogeneous



- Only a few users are very active – most are inactive
- Typically, activity (number of likes/tweets/retweets per user) has an L-shaped form
 - Use log-log plot to visualize (plots on the left)
 - Indicative of high inequality of activity



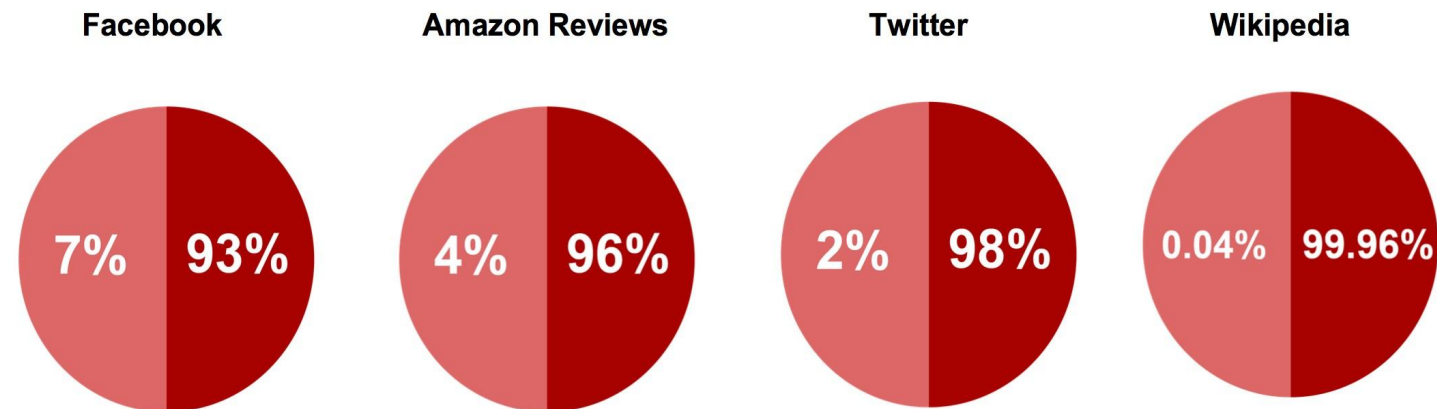
Activity Bias, or Wisdom of a Few

Link bias

- half of the Twitter users in the dataset were following only a few select celebrities (0.05% of the most popular people attracted almost 50% of all participants);
- Inequality of popularity (similar to wealth inequality)

Activity bias

How many users produce most of the content?





“digital desert”

- Activity bias generates a “digital desert” across the Web, or Web content no one ever sees.
- 1.1% of the tweets were written by people without followers.
- 31% of the Wikipedia articles added or modified in May 2014 were never visited in June.
- The size of the digital desert on the Web is in the 1% to 31% range.



Is bias bad?

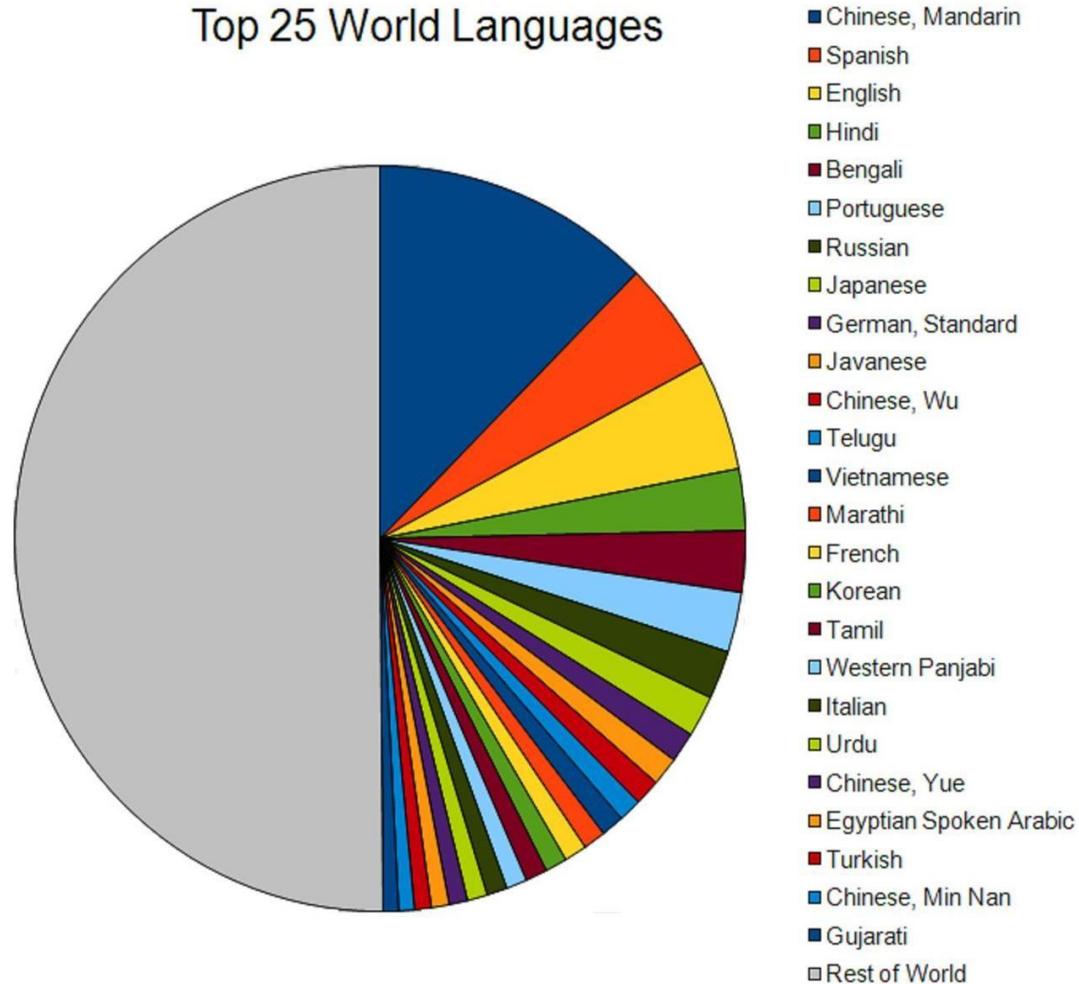
The Web works thanks to bias

- Web traffic
 - Local caching
 - Proxy/Akamai caching
- Search engines
 - Answer caching
 - Essential web pages
 - 25% queries can be answered with less than 1% of the URLs [Baeze-Yates, Boldi, Chierichetti, WWW 2015]
- E-commerce
 - Most revenue comes from few items



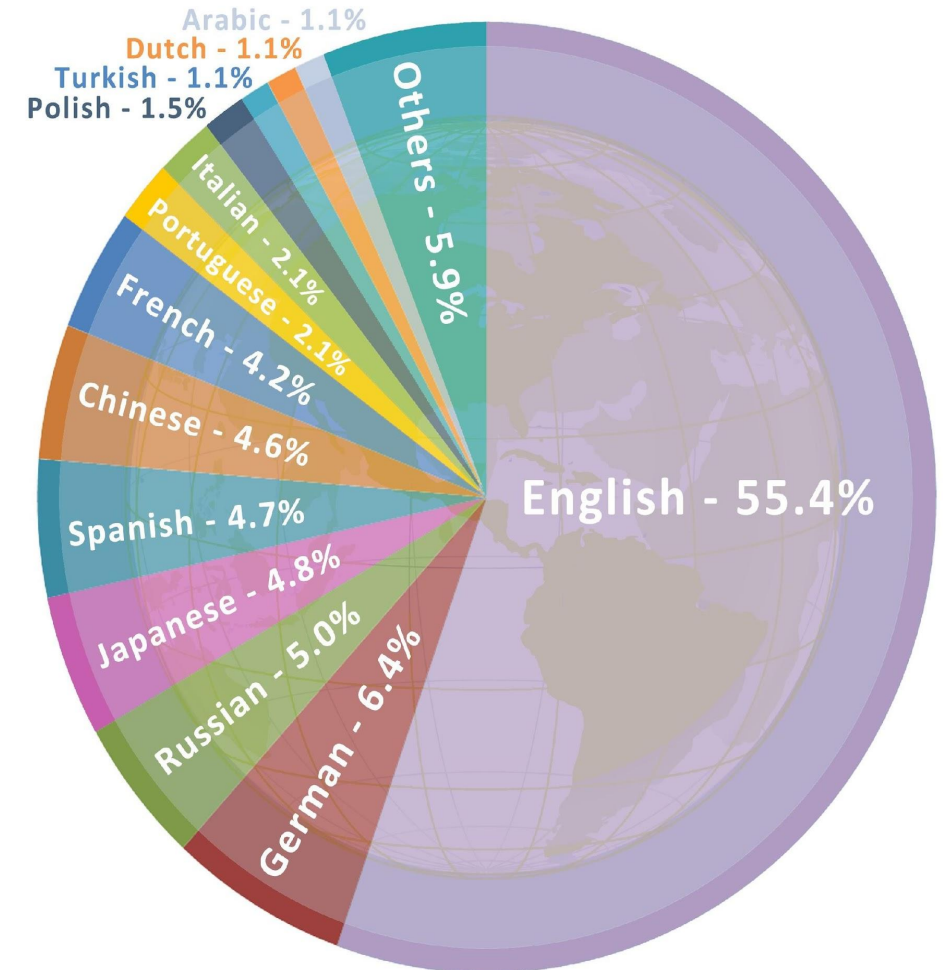
Linguistic bias

Top 25 World Languages



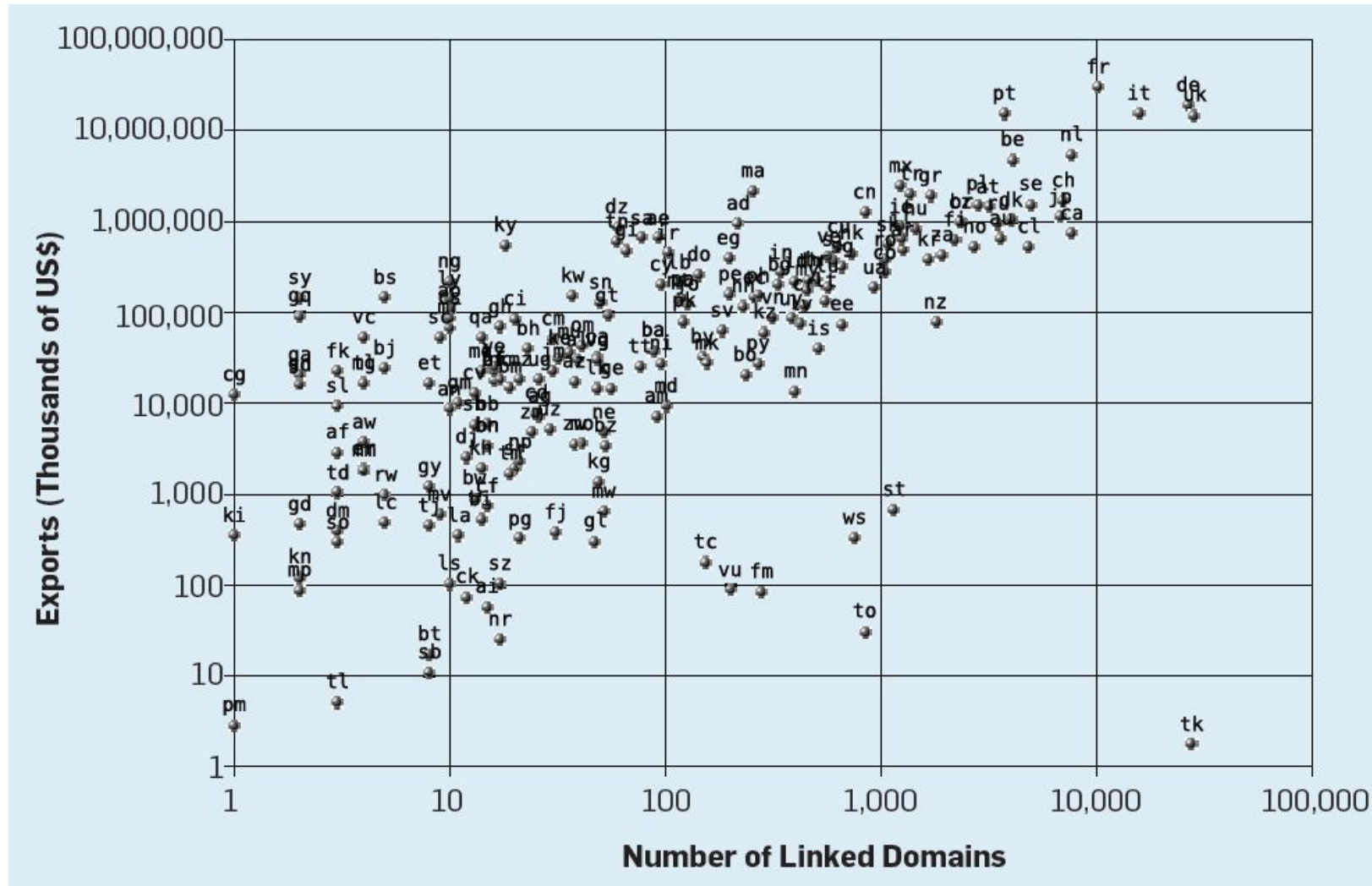
The Languages of Web Content

The percentage of the top 1 million websites available in various languages

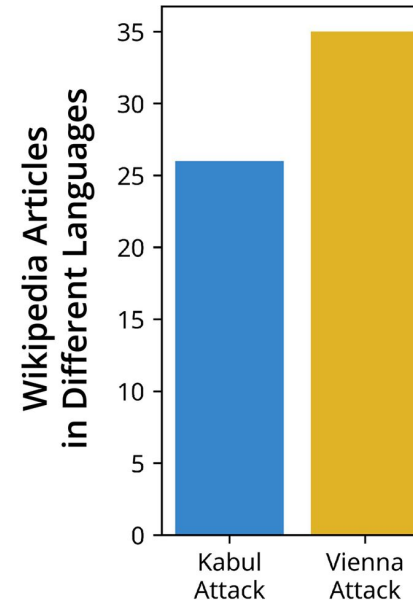
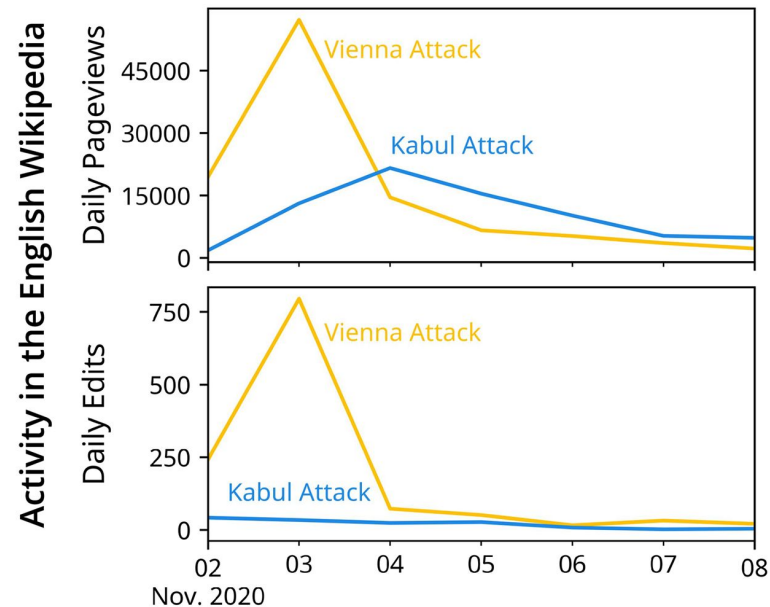




Geographic bias



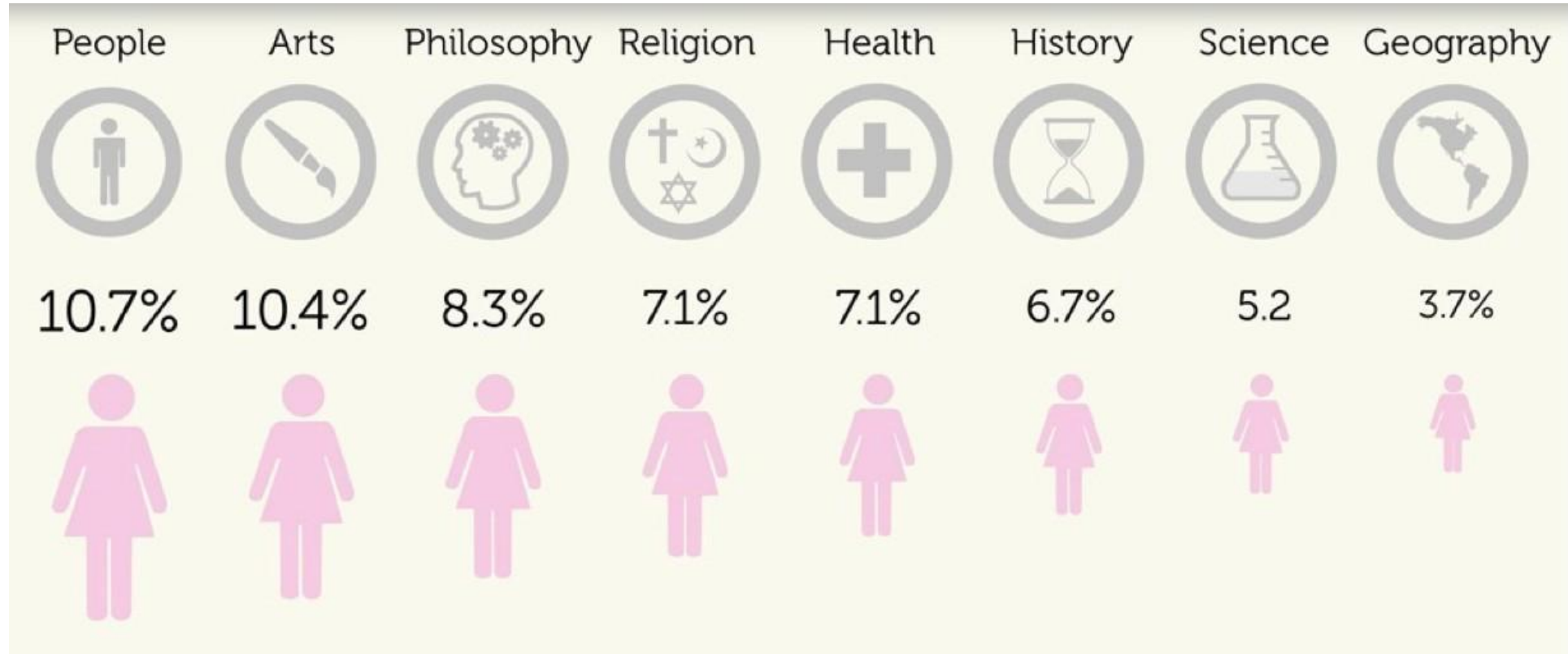
Wealth and regional disparities in attention to events on Wikipedia



Ruprecht T, Burghardt K, Helic D (2023) Poor attention: The wealth and regional gaps in event attention and coverage on Wikipedia. PLoS ONE 18(11): e0289325. <https://doi.org/10.1371/journal.pone.0289325>



Gender bias in Wikipedia



bias in
Wikipedia
biographies

[E. Graells-Garrido et al., “First Women, Second Sex: Gender Bias in Wikipedia”, ACM Hypertext’15]



Interaction between biases in content creation

- Systematic gender bias through history
- But, content creation bias too
 - less than 12% of Wikipedia biography editors are women
 - Worse gender bias in other categories (e.g., 4% of geography editors are women)
- 70% of influential journalists in the U.S. are men,
- Algorithms learning from news articles are thus learning from texts with systemic gender bias.
 - There could be additional cultural and cognitive biases
- Feedback loops amplify bias!



Platforms & biases

- User interactions on websites will be biased to what is shown
- In search systems top ranked results will get more clicks than other results
 - Ranking bias
 - The top-ranked result will thus attract more clicks than the others because it is both the most relevant and also ranked in the first position.
 - To avoid ranking bias, Web search engines need to de-bias click distribution
 - Otherwise, the popular pages become even more popular.
 - Interaction bias
- In recommender systems, items recommended will get more clicks than items not recommended
 - Vast amount of content not shown to users
- ...



Biases in the user interface

Related Searches: [tennis racket](#), [tennis shoes](#).

Shop by Category



Tennis Equipment



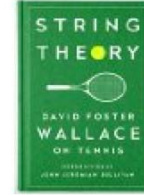
Tennis Games



Kids' Sports



Clothing, Shoes &
Jewelry



Tennis - Books

Position bias

Ranking bias

Sponsored ⓘ **Presentation bias**



Tennis Elbow Brace with Gel Comp...

\$24.50 ✓Prime

★★★★★ 7



DIMANKA Professional Table Tenni...

\$34.99

★★★★★ 9



Gamma Quick Kids 78 Ball (12 Pac...

\$19.99 ✓Prime

★★★★☆ 44



Wilson Sporting Goods Championship Extra Duty Tennis Balls (1-Can)

Jun 14, 2012

by Wilson

\$2.79 ~~\$6.99~~ Add-on Item

Add to a qualifying order to get it by **Tomorrow, May 6.**

More Buying Choices

\$0.99 new (18 offers)

\$7.99 used (2 offers)

[See newer version](#)

★★★★★ 186

Sports & Outdoors: See all 60,449 items

Social bias



Best Seller

Wilson 75 Tennis Ball Pick Up Hopper

by Wilson

\$19.96 ✓Prime

Get it by **Tomorrow, May 6**

More Buying Choices

\$18.88 new (11 offers)

\$35.00 used (1 offer)

★★★★☆ 319

Product Features

Holds 75 tennis balls with a special no spill lid (Tennis Balls NOT included)

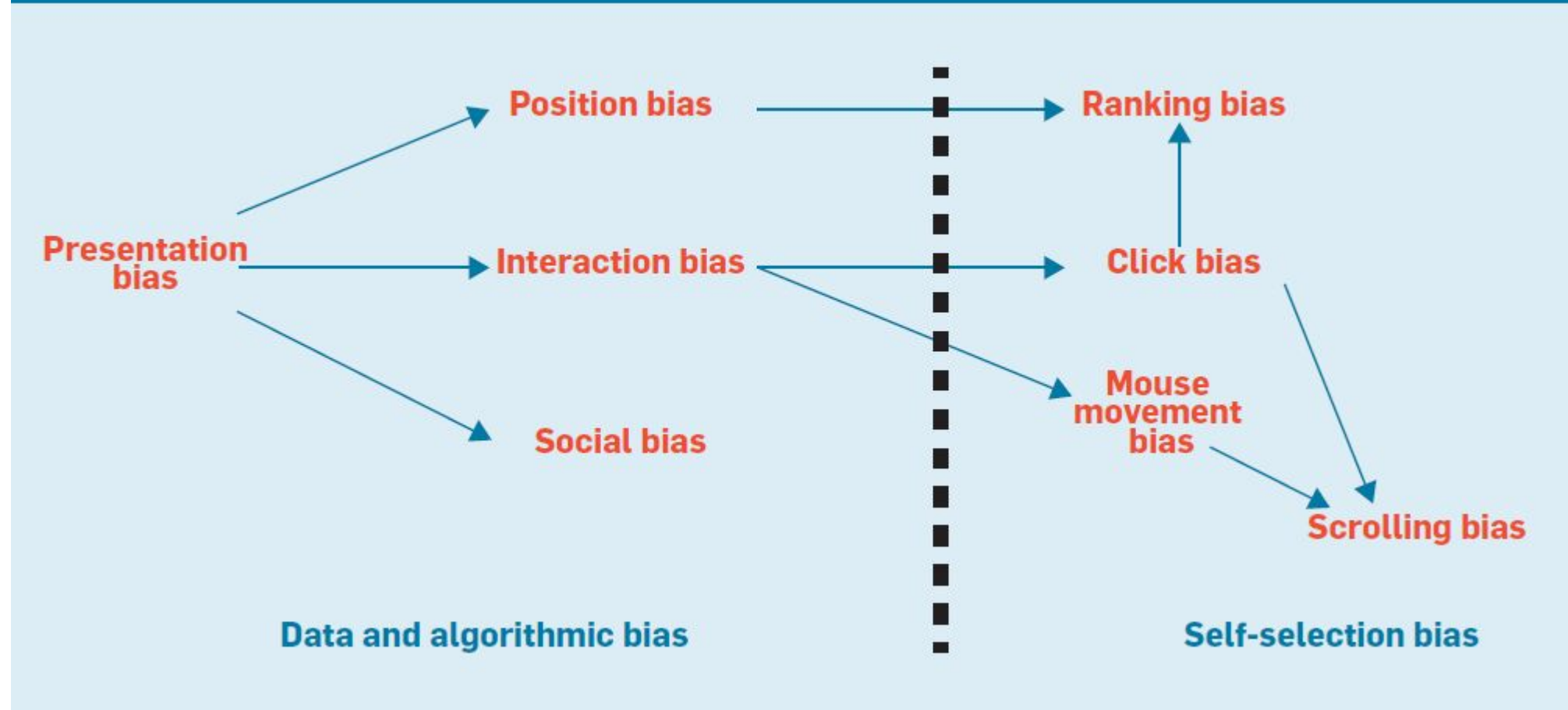
Sports & Outdoors: See all 60,449 items

Interaction bias



Cascading biases

Figure 6. Dependency graph of biases affecting user interaction.





Cascading bias: Bias begets bias

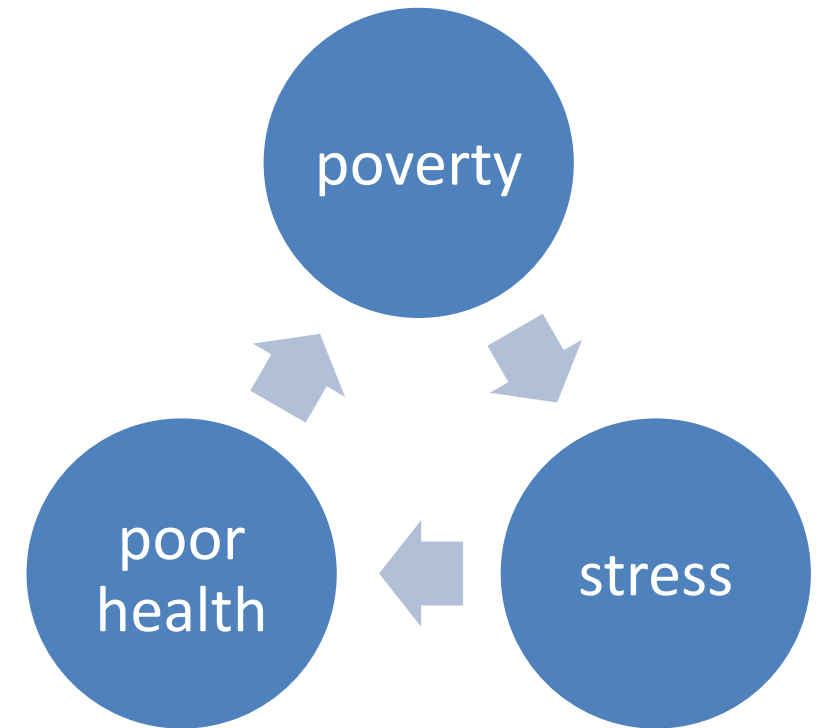
- Personalization algorithms (aka the filter-bubble effect),
 - If a personalization algorithm uses only our interaction data, we see only what we want to see, thus biasing the content to our own selection biases,
- Confirmation bias
 - We pay engage with content that is congruent with our pre-existing beliefs, feeding more biased information to personalization algorithms
- Creating feedback loops that amplify bias, a common mechanism across domains



A general mechanism for bias amplification

- **Positive feedback loops:** Advantage begets more advantage
- Cumulative advantage in science [[Merton 1968](#)]: Patterns of recognition are skewed in favor of the privileged
 - aka “the rich get richer and the poor get poorer”
 - “eminent scientists get disproportionately great credit for their contributions to science while relatively unknown scientists tend to get disproportionately little credit for comparable contributions.”
- Also see preferential attachment in networks: a few nodes get most of the links [Barabasi & Albert 1999]

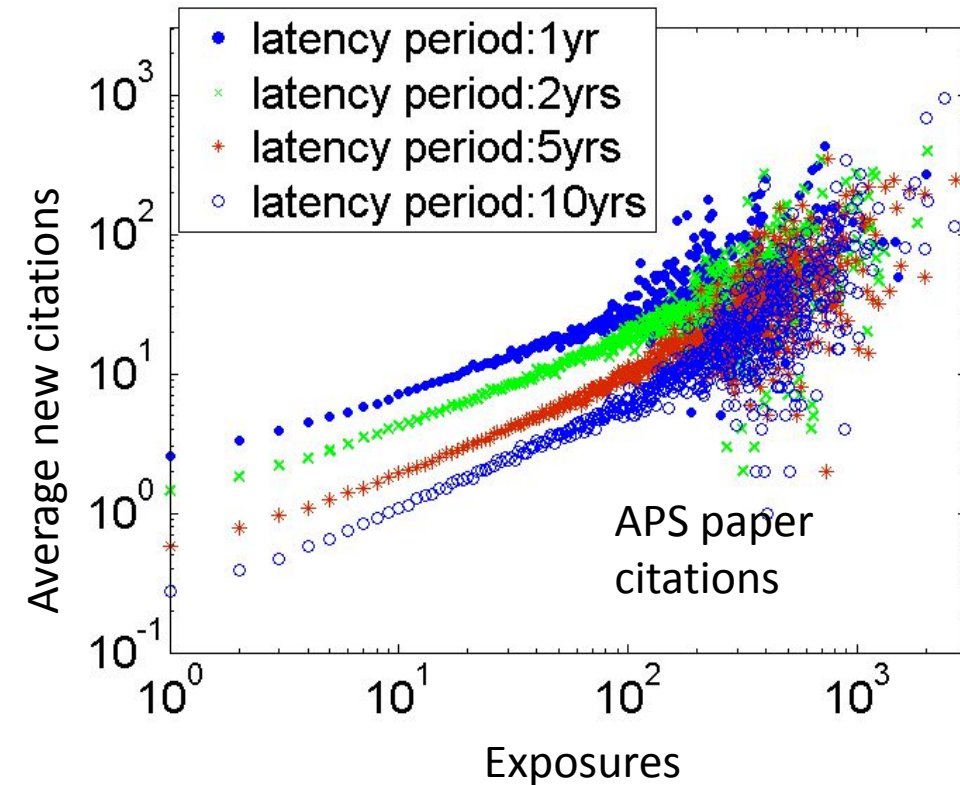
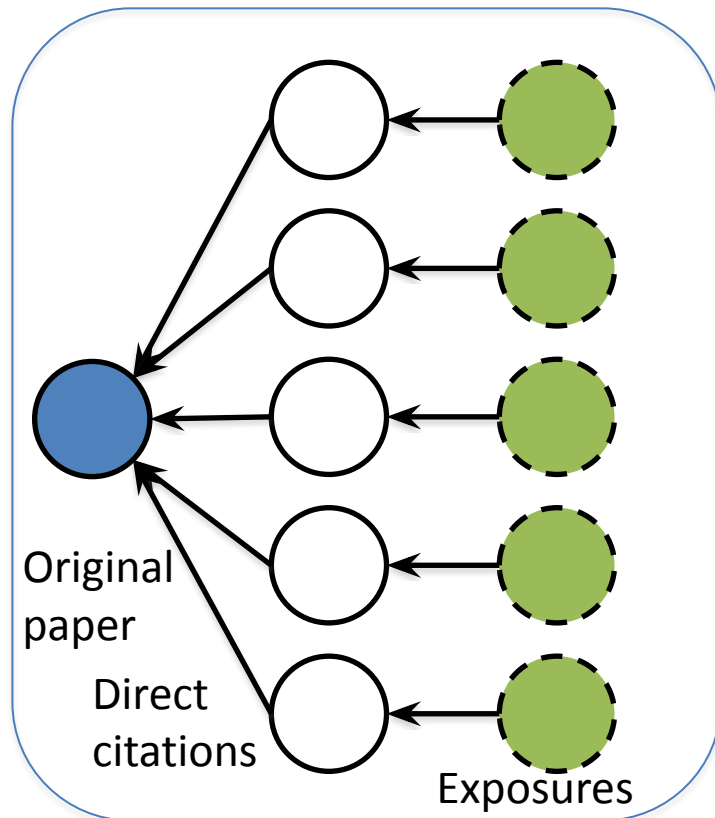
Negative feedback loops:
Disadvantage begets more disadvantage





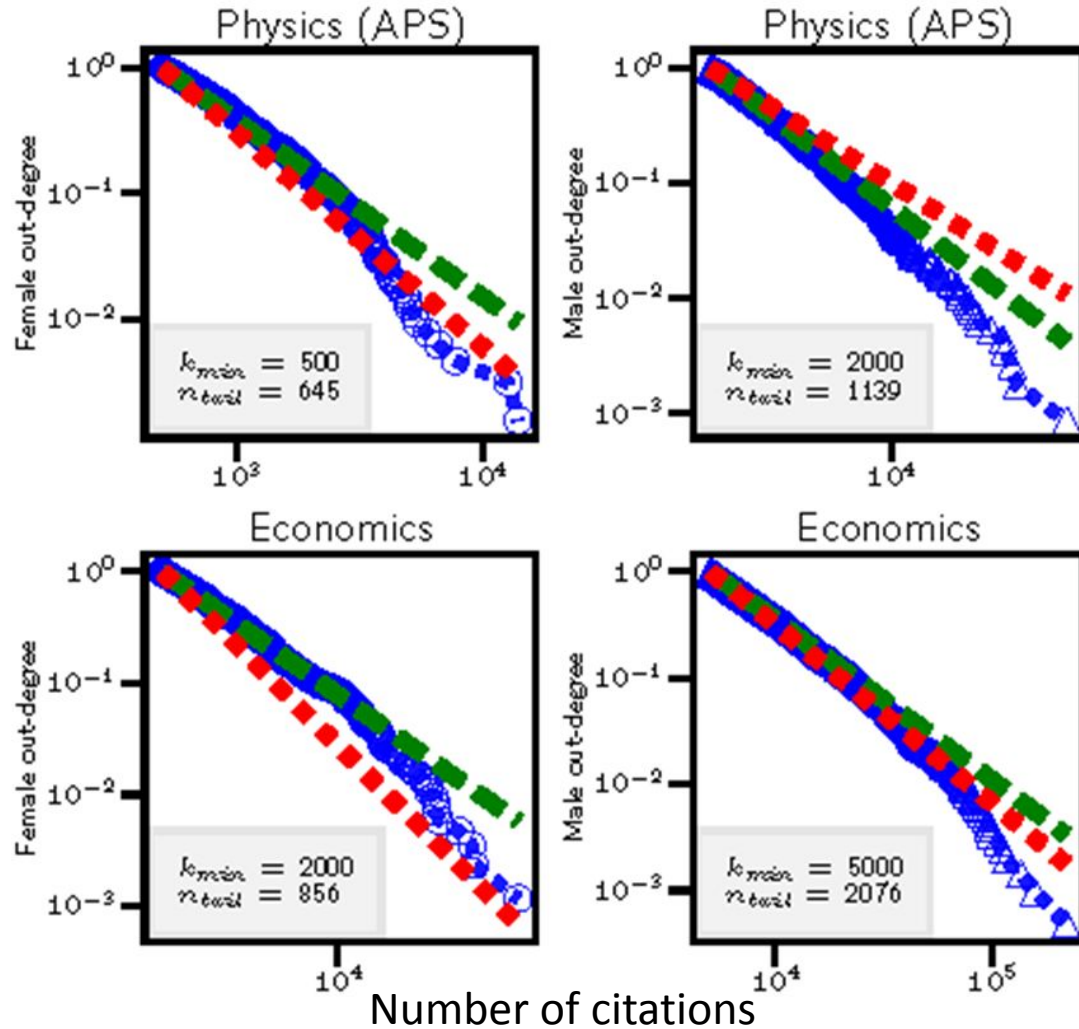
Cumulative advantage in science: citations

Citations beget more citations: each citation of a paper exposes it to a wider audience, creating opportunities for others to find and cite it in their work.





Inequality of citations

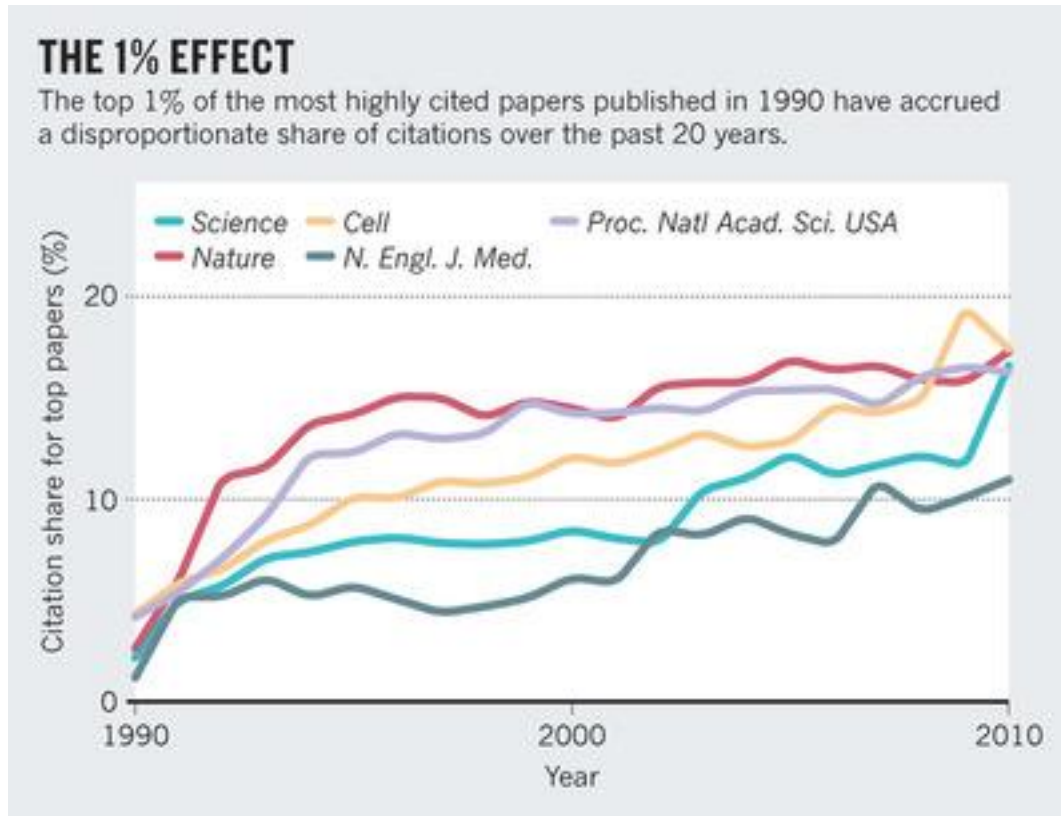


- The number of citations is highly skewed
 - Most researchers have only a few citations
 - A small number of researchers have large number of citations
 - Gini $\sim 0.6-0.7$ (cf US wealth inequality gini ~ 0.4)
- Similar distribution for the number of followers on Twitter, the number of “likes” tweets receive, etc.



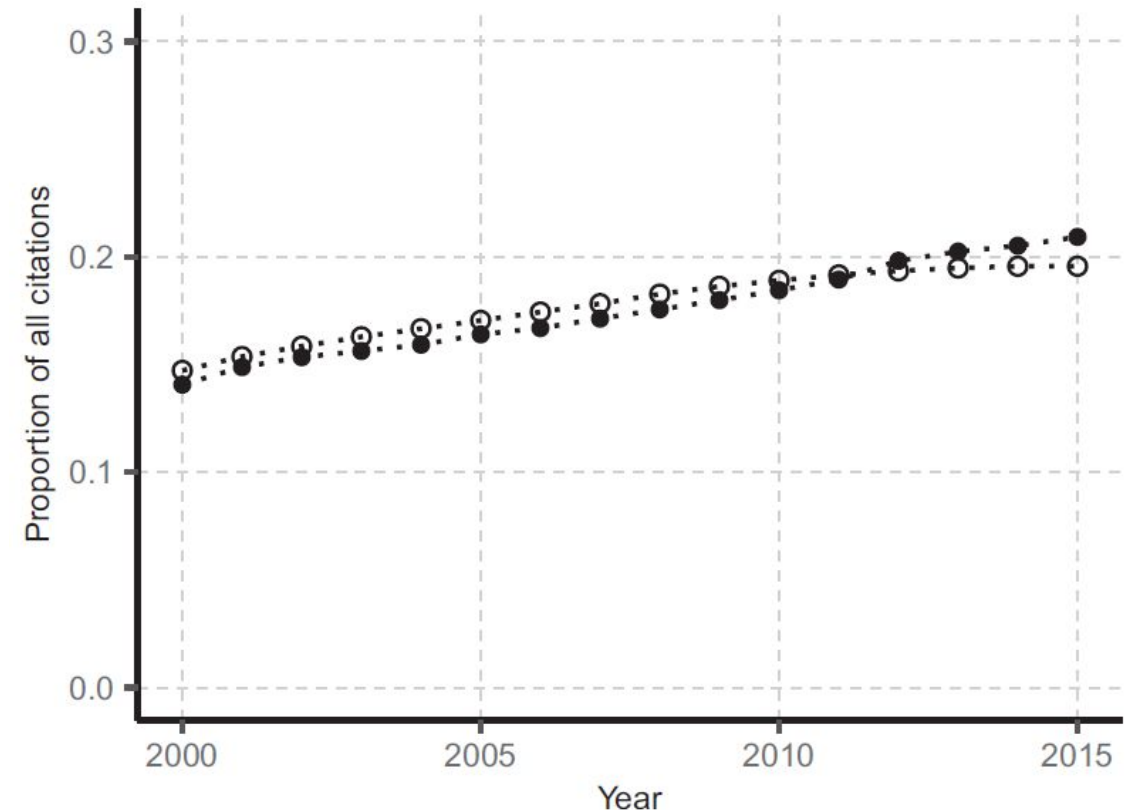
Inequality in science

A handful of papers dominates citations



Barabási, A. L., Song, C., & Wang, D. (2012). Handful of papers dominates citation. *Nature*, 491(7422), 40-40.

A handful of authors dominate citations: Share of citations going to the top-1% most-cited authors



Nielsen, M. W., & Andersen, J. P. (2021). Global citation inequality is on the rise. *Proceedings of the National Academy of Sciences*, 118(7).



COGNITIVE BIASES



Bounded rationality (aka “thinking is hard”)



Herbert A. Simon

- **Bounded rationality**
- Constraints of available time, information, and cognitive capacity limit human ability to make rational decisions

[Simon (1957). "A Behavioral Model of Rational Choice", in Mathematical Essays on Rational Human Behavior in a Social Setting.]



Daniel Kahneman



Amos
Tversky

- **Heuristics and biases**
- Mental shortcuts that help people make quick, but less accurate decisions, by focusing brain's limited resources on the most salient information

[Tversky and Kahneman (1974). Judgment under uncertainty: Heuristics and biases. *Science*
Kahneman (2011) *Thinking Fast and Slow*.]



How do individual biases affect online behavior



COGNITIVE BIAS CHEAT SHEET BECAUSE THINKING IS HARD



1 TOO MUCH INFO

SO ONLY NOTICE...

- CHANGES
- BIZARRENES
- REPETITION
- CONFIRMATION



2 NOT ENOUGH MEANING

SO FILL IN GAPS WITH...

- PATTERNS
- GENERALITIES
- BENEFIT OF DOUBT
- EASIER PROBLEMS
- OUR CURRENT MINDSET



3 NOT ENOUGH TIME

SO ASSUME...

- WE'RE RIGHT
- WE CAN DO THIS
- NEAREST THING IS BEST
- FINISH WHAT'S STARTED
- KEEP OPTIONS OPEN
- EASIER IS BETTER



4 NOT ENOUGH MEMORY

SO SAVE SPACE BY...

- EDITING MEMORIES DOWN
- GENERALIZING
- KEEPING AN EXAMPLE
- USING EXTERNAL MEMORY

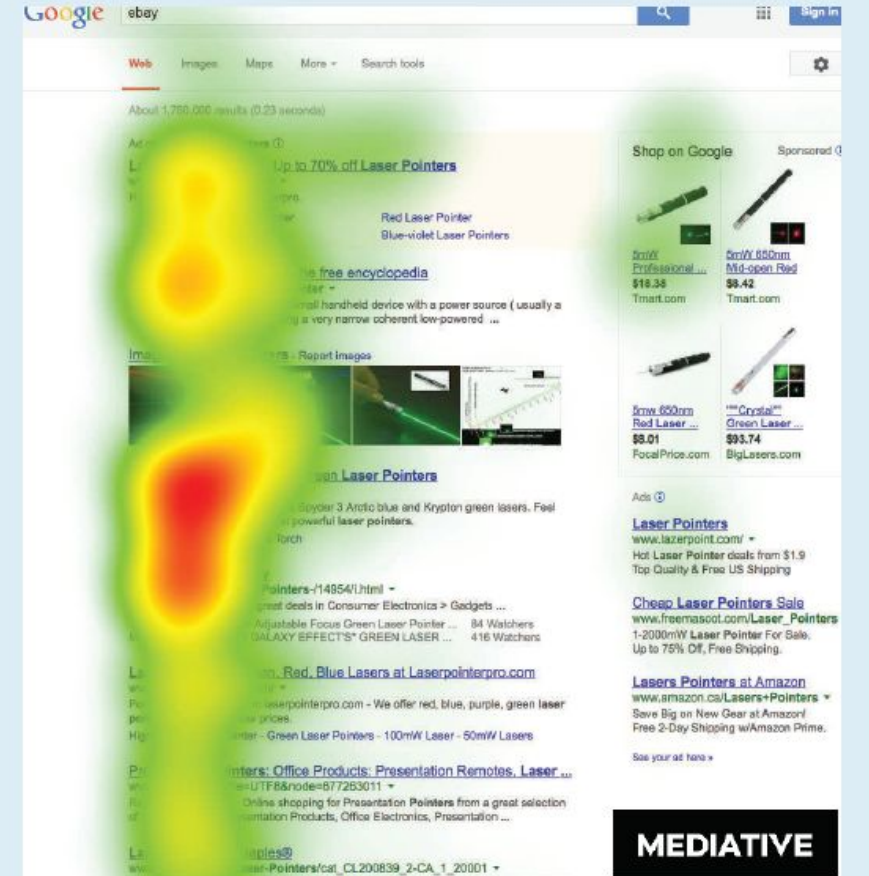
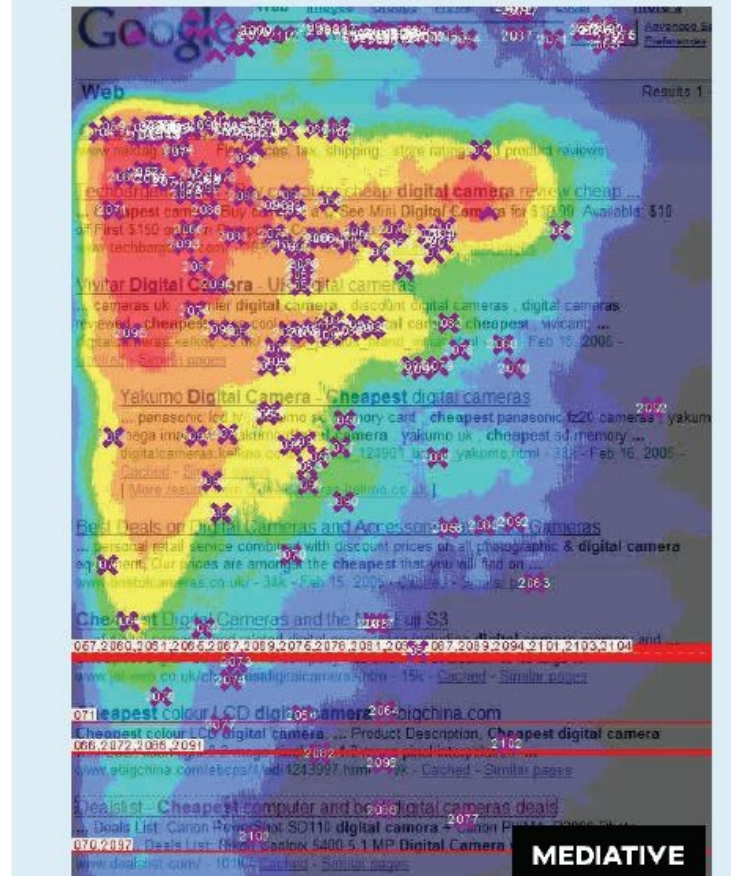
BY @BUSTER
[HTTP://BIT.LY/THINKING-IS-HARD](http://bit.ly/thinking-is-hard)

Content near the top or near images attracts attention



Figure 5. Heat maps of eye-tracking analysis on web-search results pages, from 2005 (left) to 2014 (right).¹⁸

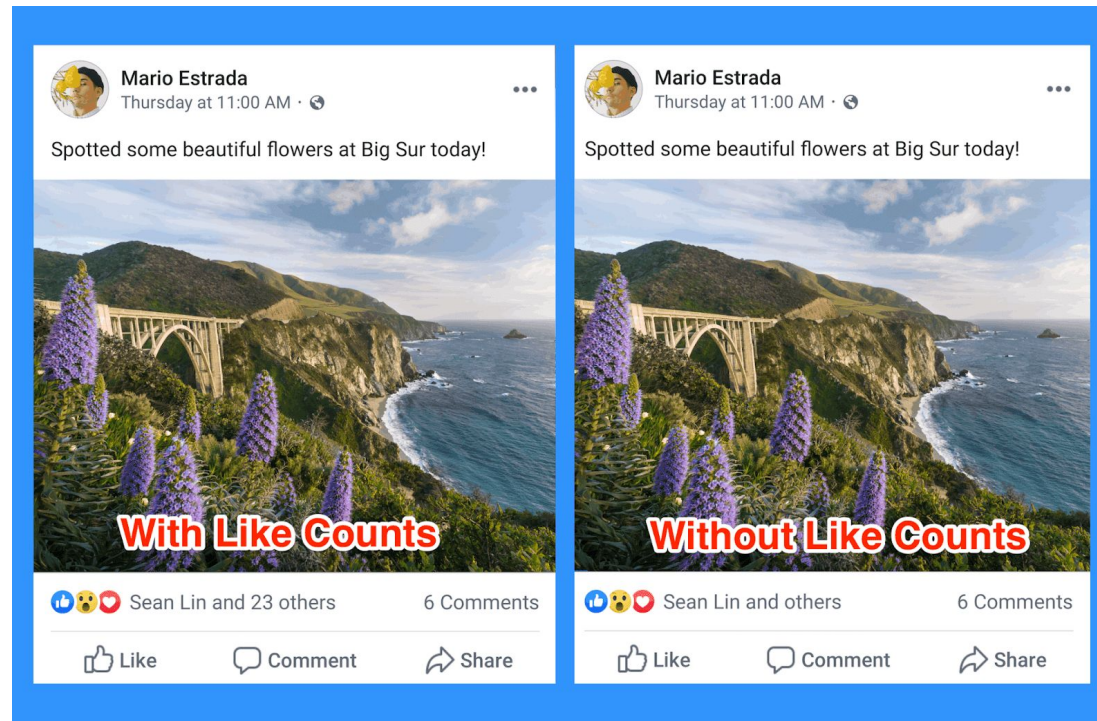
Position bias: People pay more attention to items at the top of the screen or a list of items [Payne 1951]





Social proof bias

- Signals from other people affect our judgment.
 - “social conformity,” “the herding effect”, or “the bandwagon effect”





Negativity bias: Bad is stronger than good*



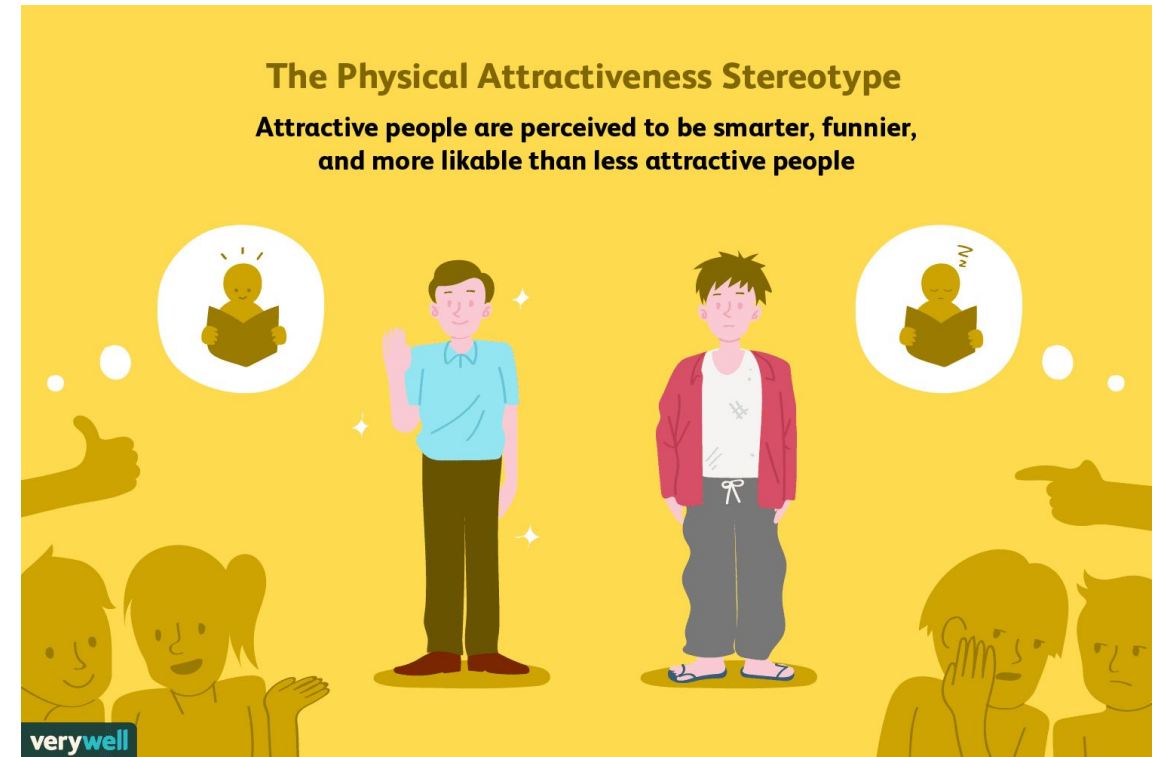
- Asymmetry in human perception
 - Criticism affects us more than praise; Negative first impressions are harder to overcome
 - Negative emotions have stronger impact than positive emotions
 - News coverage is mostly negative
- Negatively-biased credulity
 - Negative information (e.g., about threats) is more likely to be believed
 - Associated with ideology; conservatives more likely to believe negative information

* Baumeister, R. F., Bratslavsky, E., Finkenauer, C., & Vohs, K. D. (2001). Bad is stronger than good. *Review of general psychology*, 5(4), 323-370.



Halo effect

- Tendency for positive impressions of a person in one area to positively or negatively influence one's opinion or feelings in other areas.
- One trait of a person or a thing is used to make an overall judgment about them





INTERACTIONS BETWEEN COGNITIVE BIASES & ALGORITHMS



Crowdsourced ranking

- **Crowdsourced ranking**

- aggregates decisions of many people to rank items

- “quality items float to the top”

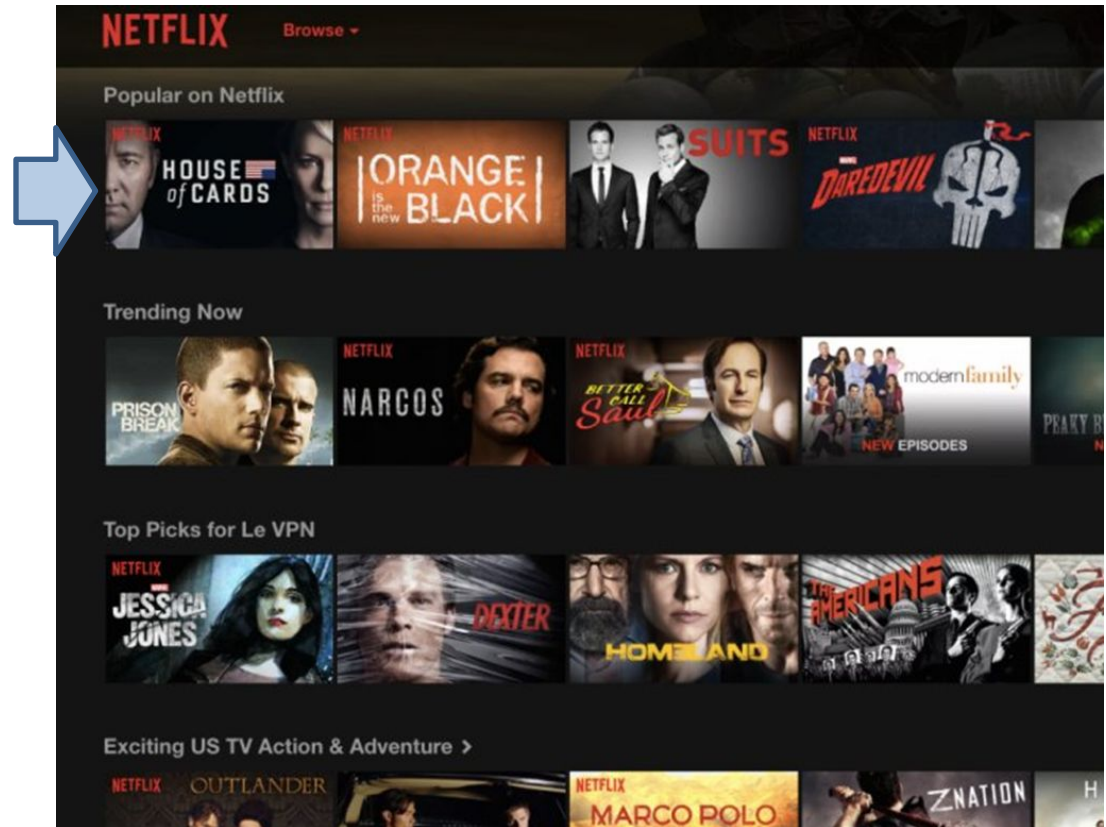
- **Instability in popularity ranking**

- Music Lab study [Salganik+ 2006]

- Few items become hits; most languish in obscurity

- Can't predict what items become hits
 - Harry Potter book was rejected 9 times

Popularity ranking: Popular on Netflix (or Twitter, Reddit, StackOverflow, ...)

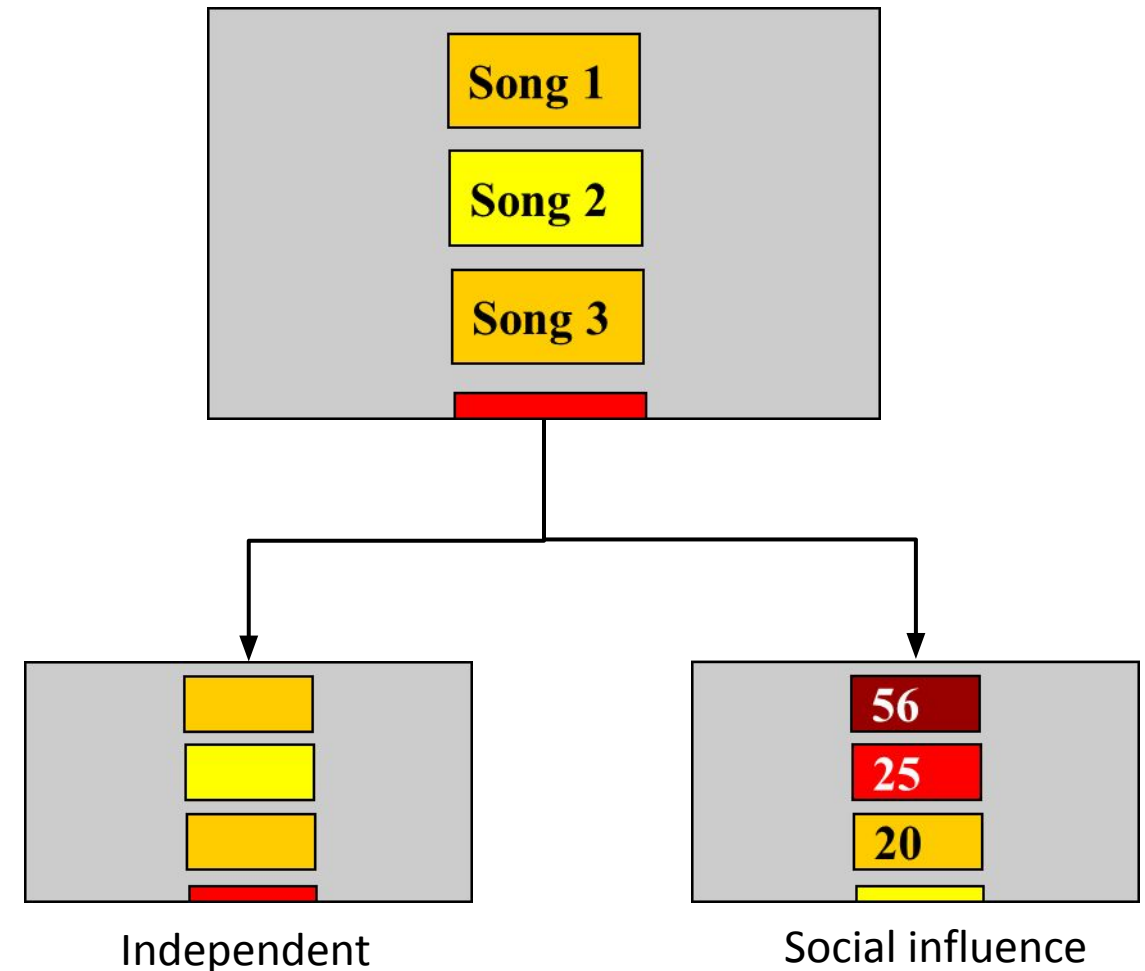




MusicLab study

Matthew J. Salganik, Peter Sheridan Dodds, Duncan J. Watts. Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market. *Science* 10 Feb 2006 : 854-856

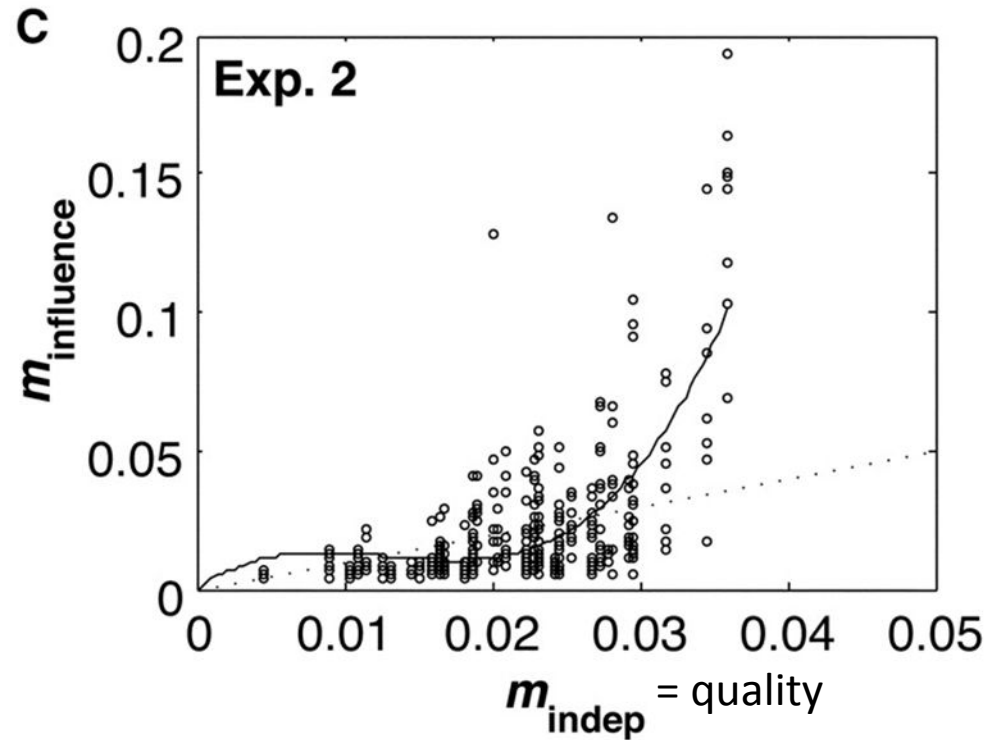
- Online experiment to simulate a cultural market
- Users shown a list of songs in one of two conditions
 - **Independent:** songs in random order
 - **Social influence:** songs ranked by popularity
- Users download songs they like
 - Popularity = fraction of all downloads a song receives
 - Quality = popularity in the independent condition





Popularity vs quality

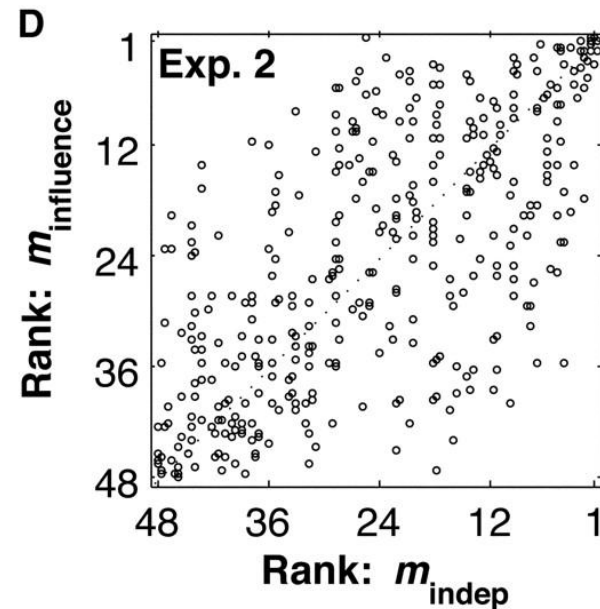
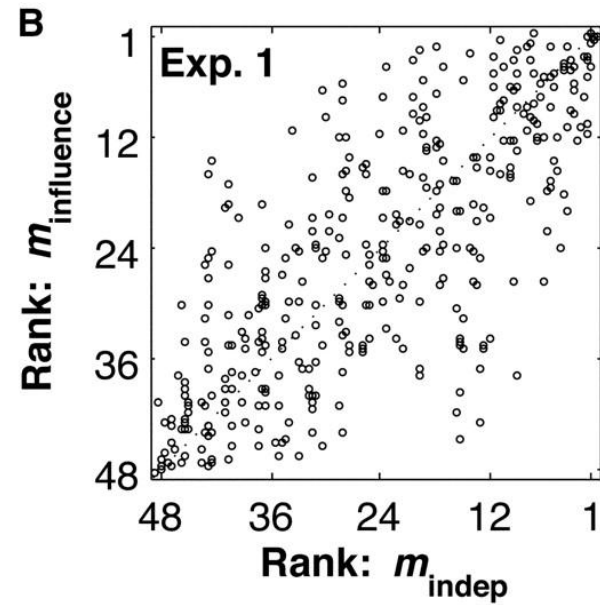
Song market share*: social influence vs independent conditions



* market share = fraction of all downloads in an experiment attributed to a given song

Ranking instability

- Huge differences in popularity (inequality)
 - Few items become very popular
- Popularity is weakly correlated with quality
 - The best and worst songs were correctly ranked.
 - Rank of mediocre songs is unpredictable
- Multiple worlds experiment
 - Rerun the experiment from the same initial conditions. Different songs become popular

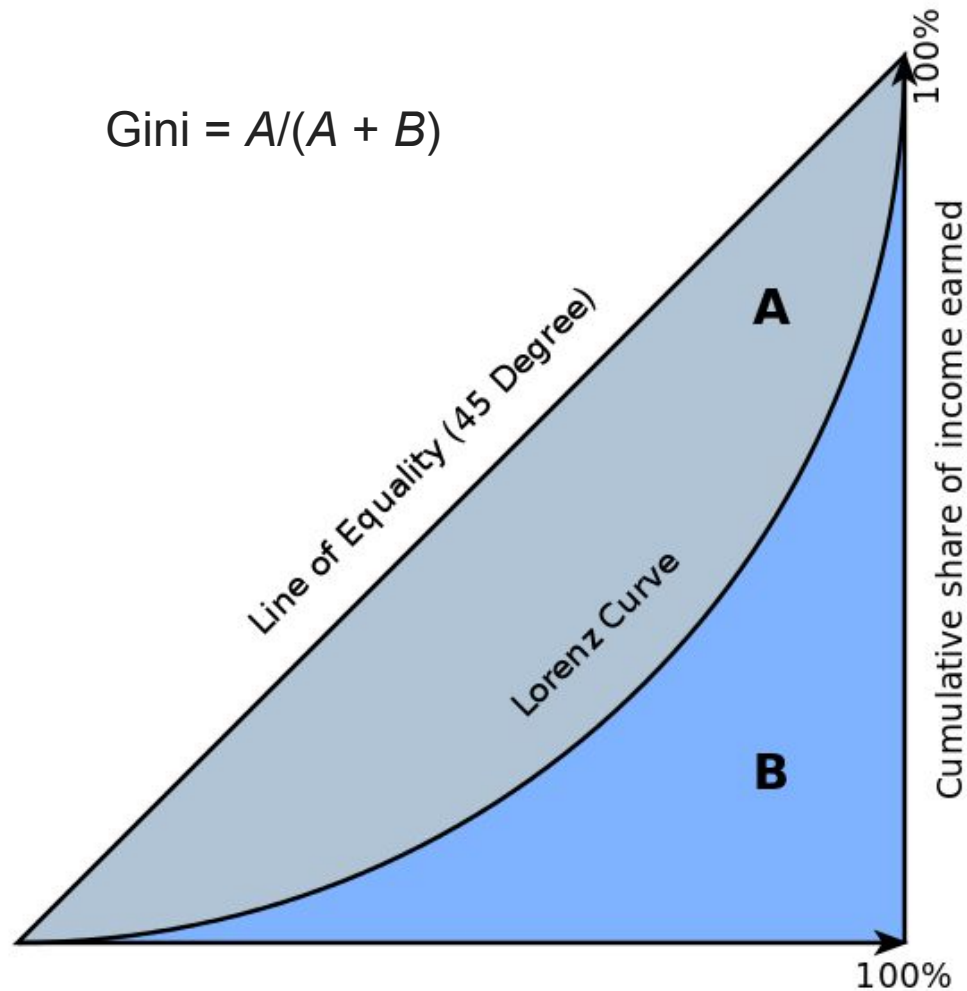




Measuring inequality

Gini coefficient measures the degree of inequality in a distribution of values.

$$\text{Gini} = A / (A + B)$$

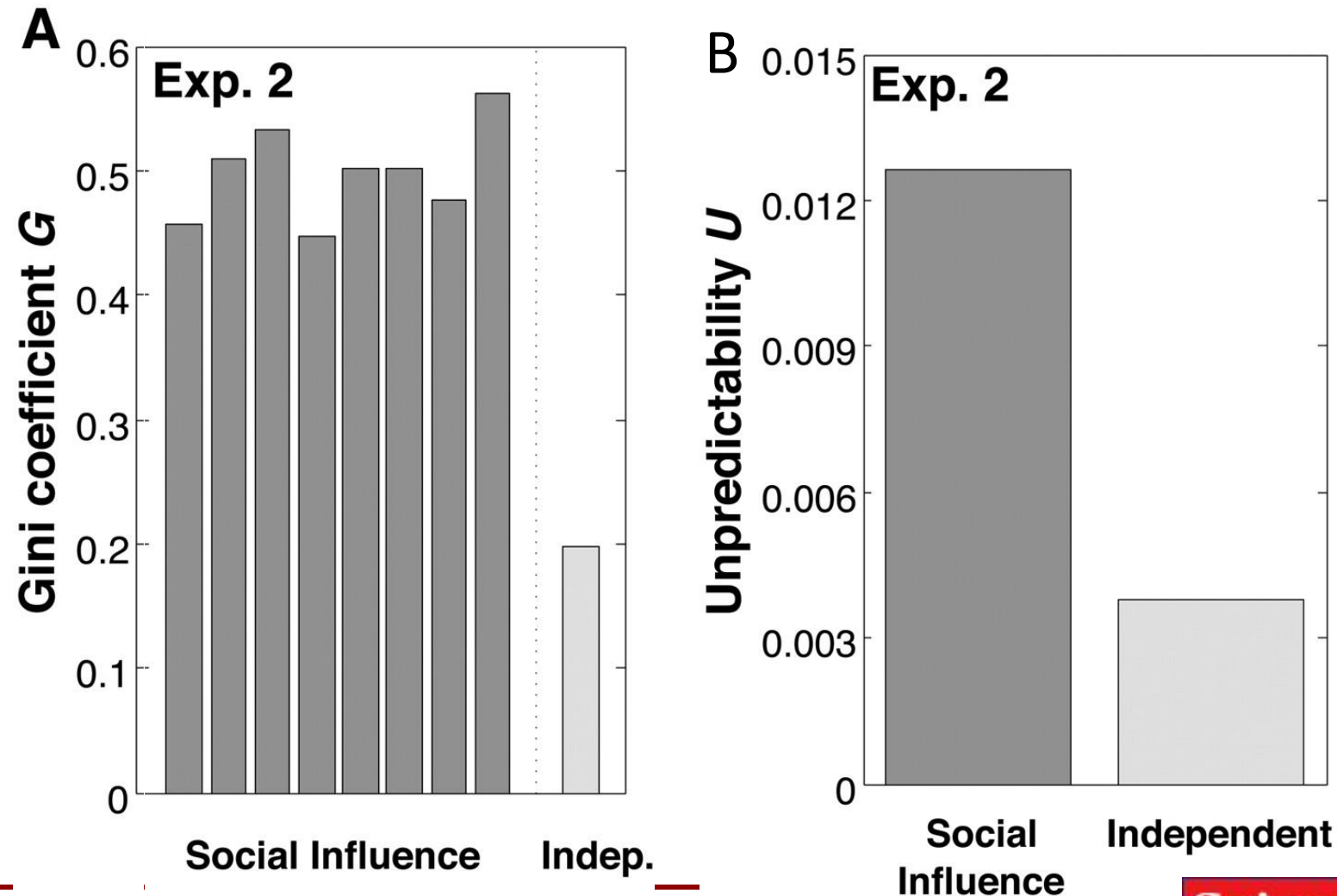


$$G = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2 \sum_{i=1}^n \sum_{j=1}^n x_j}$$

/r Cumulative share of people from lowest to highest incomes



Inequality (A) and unpredictability (B) of song market share in the social influence vs independent conditions worlds





Conclusions of the MusicLab study

- Social influence distorts collective outcomes in markets
 - Popularity is only weakly correlated with quality
 - Social influence is responsible for the inequality and unpredictability of popularity

But, MusicLab study did not attempt to control for the salience of the social signal – attention paid to songs solely due to their position in the list



Social influence or position bias?

- **Social influence:** People pay more attention to the popular choices
- **Position bias:** People pay more attention to items at the top of the screen or a list of items [Payne 1951]



[Buscher et al, CHI'09]



Measuring position bias

- Quantify the effects of position bias on individual's choices and collective outcomes (story popularity) via Amazon Mechanical Turk experiments

No social influence condition

Finish

Please recommend science topics from the list below

 recommend	Grab the Leash: Dog Walkers More Likely to Reach Exercise Benchmarks Man's best friend may provide more than just faithful companionship: A new study led by a Michigan State University researcher shows people who owned and walked their dogs were 34 percent more likely to meet federal benchmarks on physical activity.
 recommend	No Two of Us Are Alike -- Even Identical Twins: Pinpointing Genetic Determinants of Schizophrenia Just like snowflakes, no two people are alike, even if they're identical twins according to new genetic research from The University of Western Ontario. Molecular geneticist Shiva Singh has been working with psychiatrist Dr. Richard O'Reilly to determine the genetic sequencing of schizophrenia using...
 recommend	Lab-Grown Meat Would 'Cut Emissions and Save Energy' Meat grown using tissue engineering techniques, so-called 'cultured meat', would generate up to 96% lower greenhouse gas emissions than conventionally produced meat, according to a new study.
 recommend	Still Counting Calories? Your Weight-Loss Plan May Be Outdated The most detailed long-term analysis of the factors that influence weight gain shows that conventional wisdom may not be the best approach.
 recommend	Happy Guys Finish Last, Says New Study On Sexual Attractiveness "Women find happy guys significantly less sexually attractive than swaggering or brooding men, according to a new University of British Columbia study that helps to explain the enduring allure of ""bad boys"" and other iconic gender types."
 recommend	Storing Water for a Dry Day Leads to Suits A small water utility in California sued to challenge the wealthy farming interests that dominate two of the country's largest water banks.
 recommend	Governor Says Montana Was Misled on Oil Spill

Social influence condition (#likes shown)

Finish

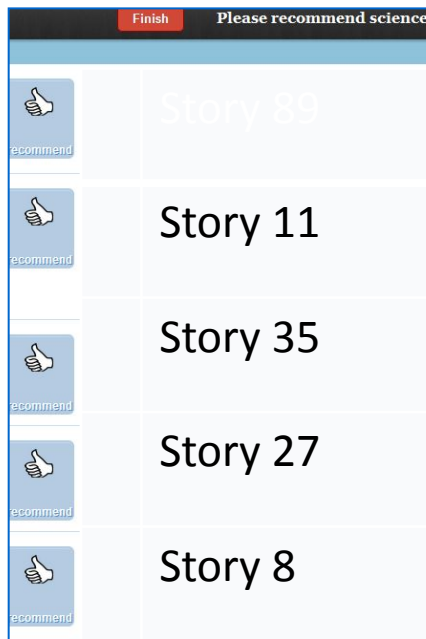
Please recommend science topics from the list below

 101 recommend	Neuroscientists Uncover Neural Mechanisms of Object Recognition Certain brain injuries can cause people to lose the ability to visually recognize objects, for example, confusing a harmonica for a cash register.
 57 recommend	A Wise Man's Treatment for Arthritis: Frankincense? The answer to treating painful arthritis could lie in an age old herbal remedy -- frankincense, according to Cardiff University scientists. Cardiff scientists have been examining the benefits of frankincense to help relieve and alleviate the symptoms of the condition.
 57 recommend	You Are What You Tweet: Tracking Public Health Trends With Twitter Twitter allows millions of social media fans to comment in 140 characters or less on anything: an actor's outlandish behavior, an earthquake's tragic toll or the great cheese sandwich.
 91 recommend	New Means of Overcoming Antiviral Resistance in Influenza Researchers from the University of California, Irvine, with assistance from the San Diego Supercomputer Center at UC San Diego, have found a new approach to the creation of customized therapies for virulent flu strains that resist current antiviral drugs.
 109 recommend	Could a Birth Control Pill for Men Be On the Horizon? Retinoic Acid Receptor Antagonist Interferes With Sperm Production Researchers at Columbia University Medical Center are honing in on the development of what may be the first non-steroidal, oral contraceptive for men. Tests of low doses of a retinoic acid receptor antagonist (RARs), whose ligands are metabolites of vitamin A, showed that...
 recommend	New Strategy to Attack Tumor-Feeding Blood Vessels

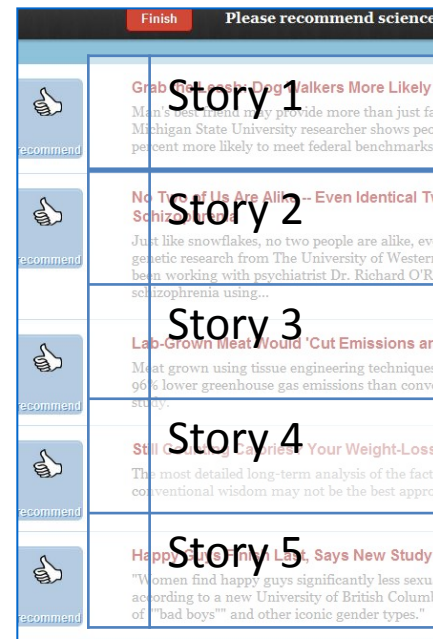


4 algorithmic ranking schemes

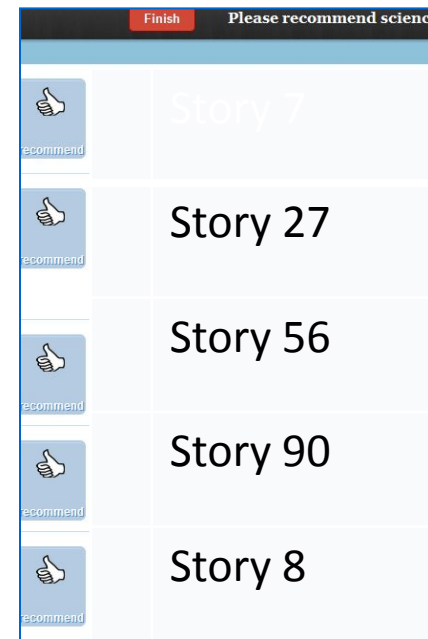
- Participants asked to recommend (like) stories from a list 100 science stories
 - click on URL to read the story
 - vote for (like) the story
- Vary order ☐ measure outcomes (popularity)



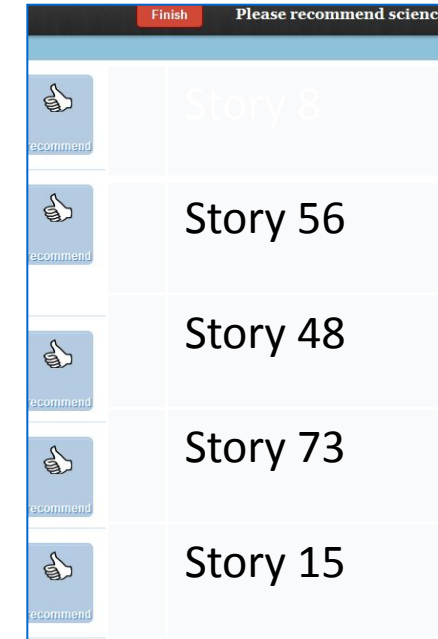
[random order]



[fixed order]



[by popularity]

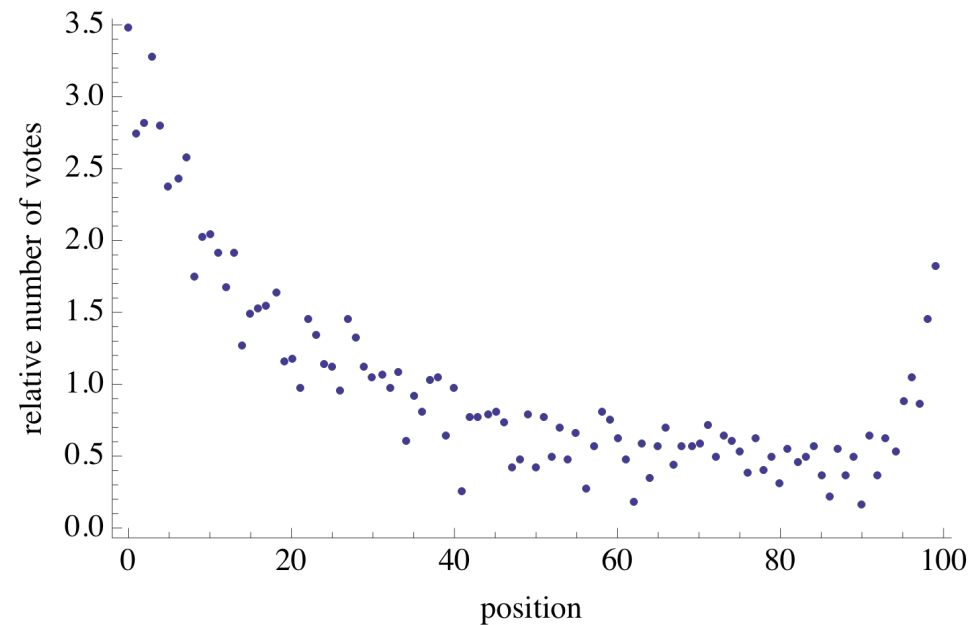


[by recency]



Position bias

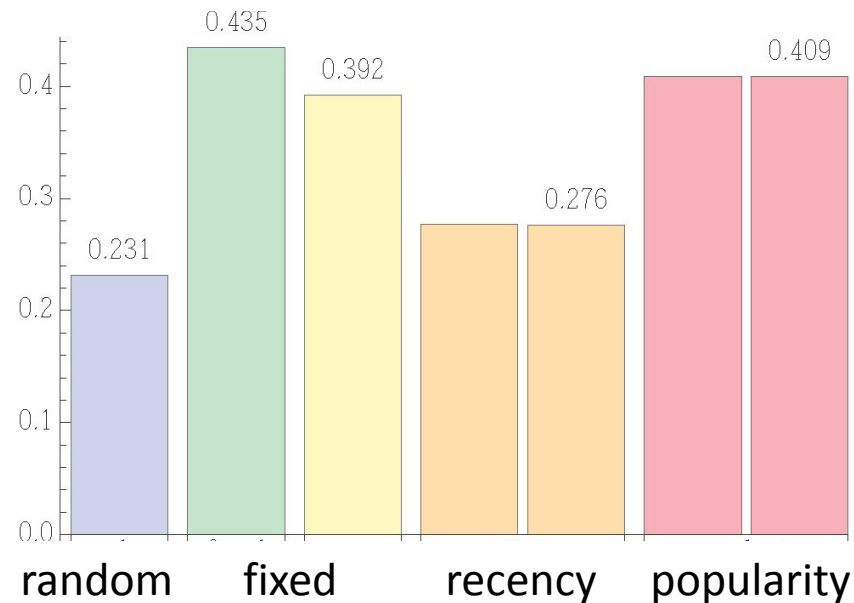
All things being equal, items in top list positions receive 4x-5x more attention than those in lower positions



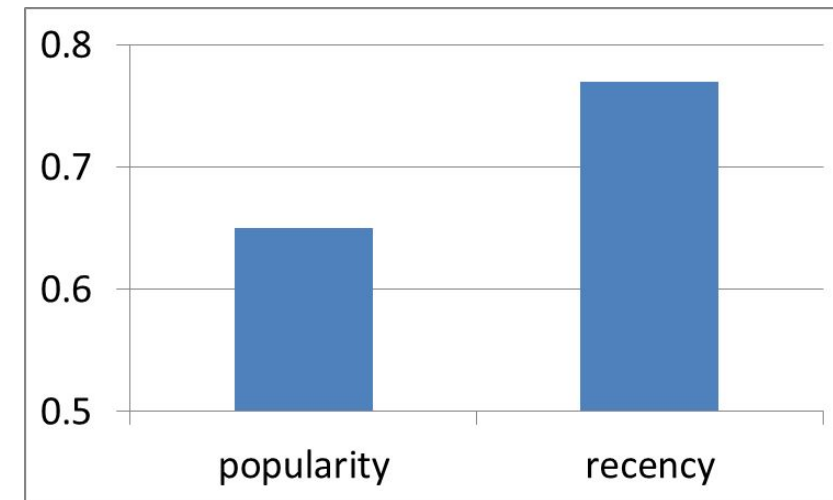


Same outcomes as MusicLab

Inequality of popularity: Gini coefficient of the number of likes received



Unpredictability of popularity: Correlation of outcomes across parallel worlds

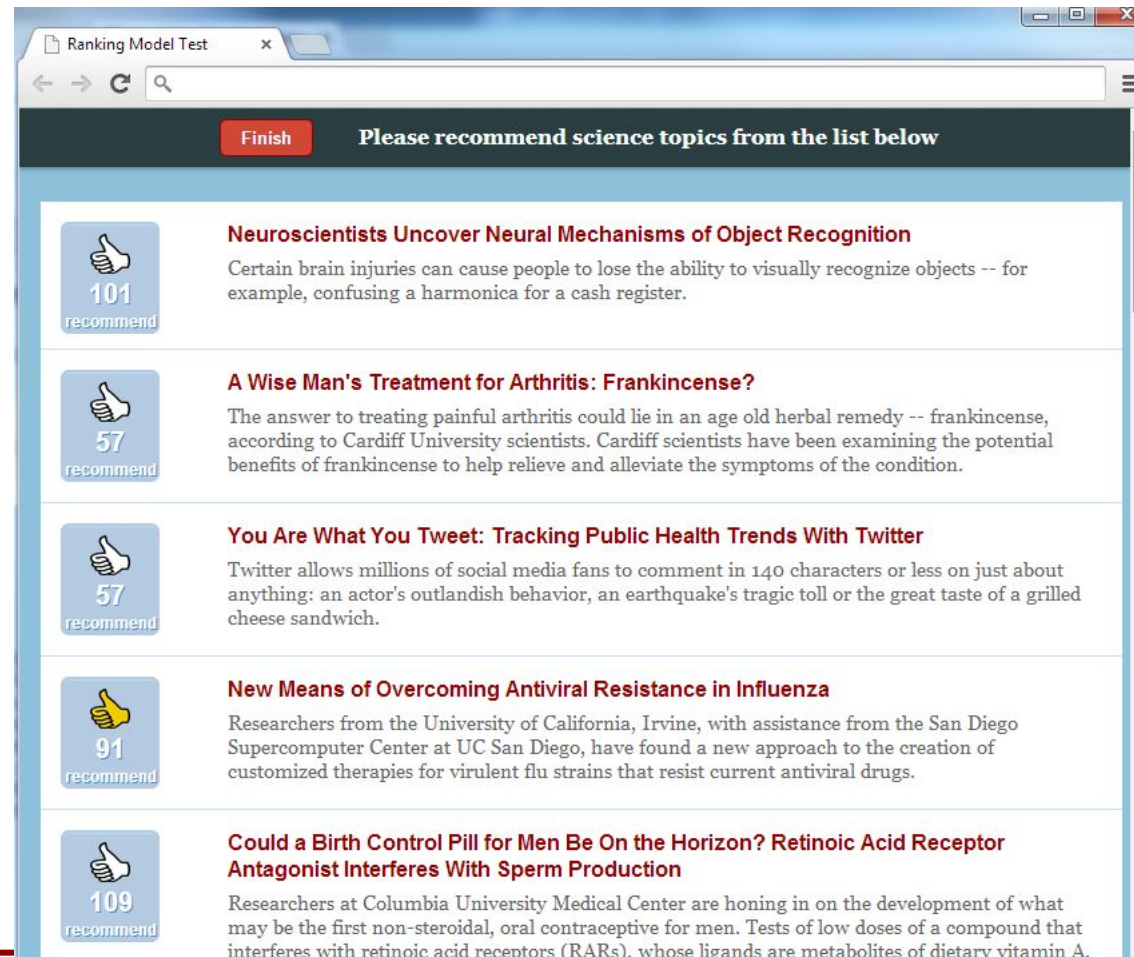


Position bias results in large inequality and unpredictability when ranking items by popularity



Measuring social influence bias

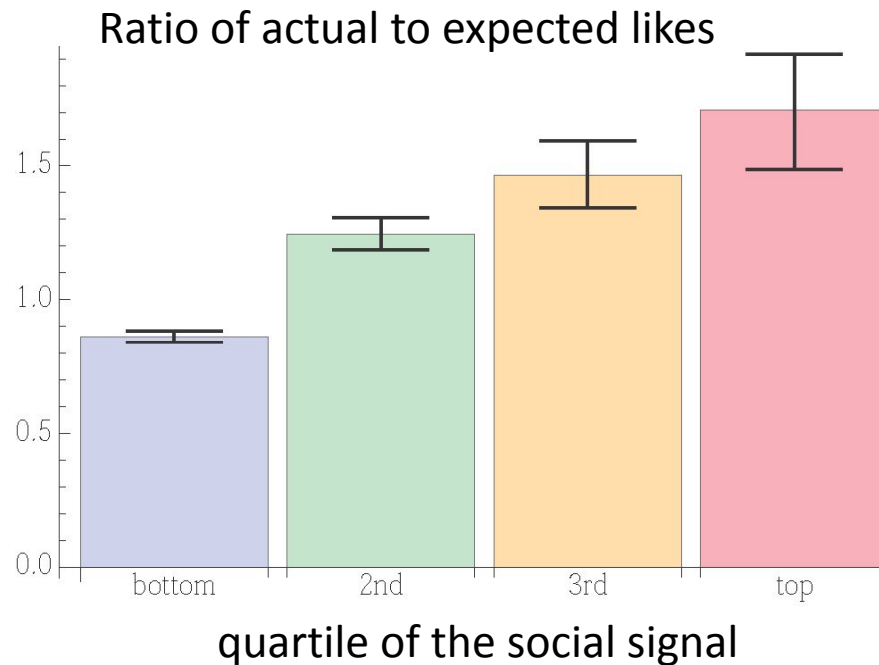
How does “social proof” (the number of likes) affect collective outcomes (popularity)?



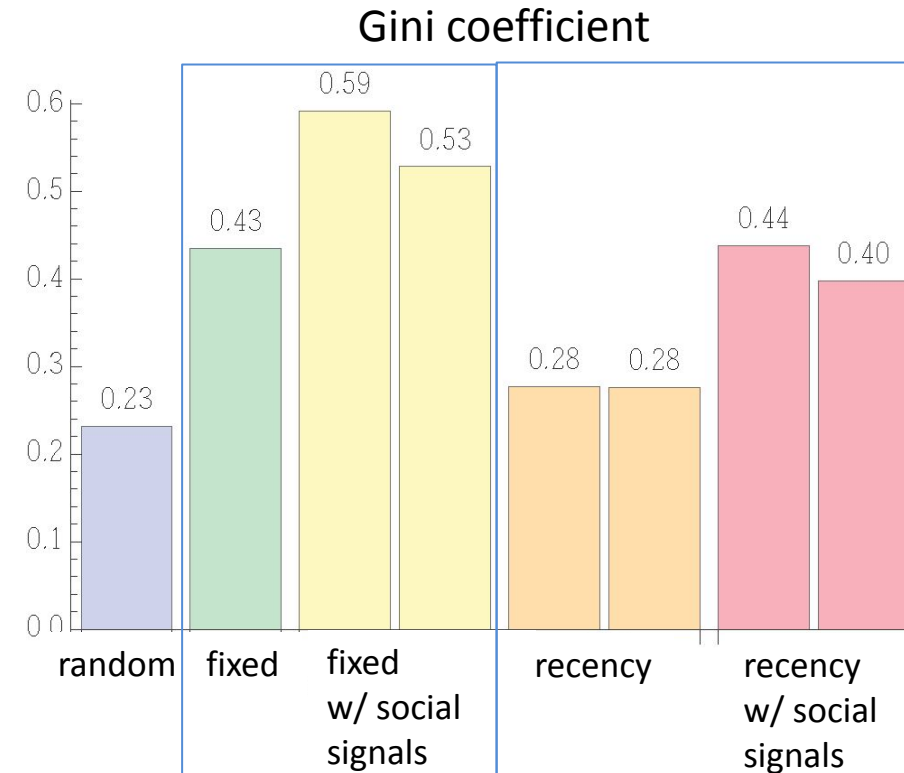


Social influence and collective outcomes

Items with higher “social proof” get more attention (after controlling for quality and position)



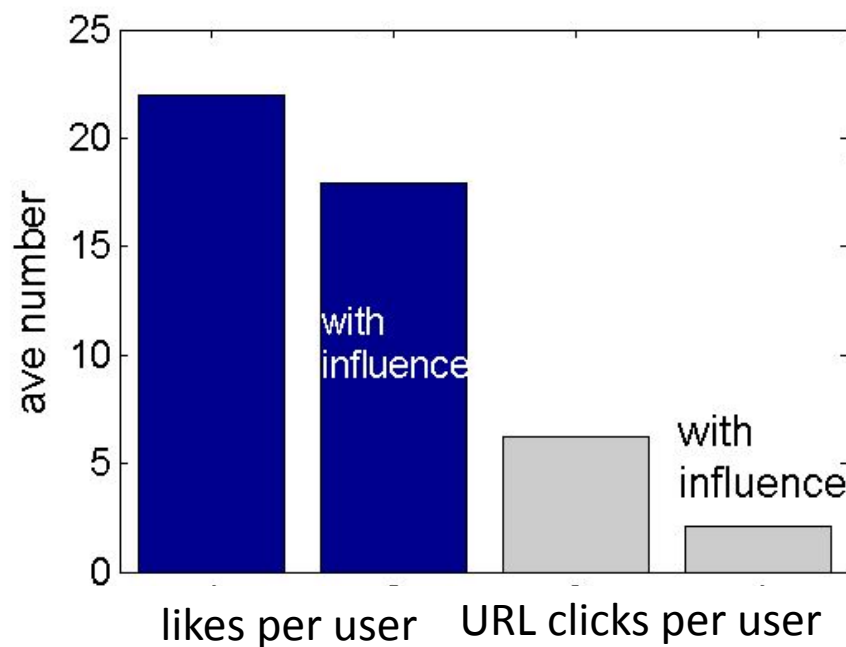
“Social proof” (influence) increases inequality



Social influence improves collective efficiency



- **Social influence reduces effort:**
people read and like fewer stories
with social signals (20% less time)





Takeaways

- **Socially-embedded AI algorithms are inherently dynamic**
 - Interactions between people and AI systems produces new data
 - Interactions are predictably skewed
 - ... often amplify bias